Session 20 · Tuesday 3 November 2026 · A home guide. The first one is worked for you; by the fourth the scaffolding is gone. Answers are released after the session.

The network solved by hand, four times

Nine reactions, six balanced metabolites. Everything below is done with the same six lines, and a solver is never needed:

REACTION		CARBON
v_{uptake}	$\longrightarrow$ G6P, capped at 10	source
v_{pgi}	G6P $\longrightarrow$ F6P	6 → 6
v_{zwf}	G6P $\longrightarrow$ P5P + CO_2	6 → 6
v_{tpi}	F6P $\longrightarrow$ 2 T3P	6 → 6
v_{lower}	T3P $\longrightarrow$ PYR + ATP	3 → 3
v_{ackA}	PYR $\longrightarrow$ Ace + CO_2 + ATP	3 → 3
v_{tca}	PYR $\longrightarrow$ 3 CO_2 + 3 ATP	3 → 3
v_{biomass}	2 T3P + P5P + 4 ATP $\longrightarrow$	sink
v_{atpm}	ATP $\longrightarrow$	sink

The only tool is $S\mathbf{v} = 0$: for every balanced metabolite, what is made equals what is used. Write b for v_{biomass} throughout.

At each transition the sheet asks *why does that step follow?* Answer in writing before you go on.

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1 · Fully worked — the wild type

P5P. One producer, v_{zwf}. One consumer, biomass, which takes one P5P per unit. So $v_{\text{zwf}} = b$.

G6P. Made at 10, used by two reactions: $10 = v_{\text{pgi}} + v_{\text{zwf}}$, so $v_{\text{pgi}} = 10 - b$.

F6P. One producer, one consumer, so $v_{\text{tpi}} = v_{\text{pgi}} = 10 - b$.

> *v_{tpi} and v_{zwf} are both lumps. v_{tpi} is phosphofructokinase, aldolase and triose phosphate isomerase on one line; the isomerase by itself only interconverts the two trioses. Worth holding on to when you read what Fong & Palsson's model predicts for the tpi deletion.*

T3P. Made at $2v_{\text{tpi}}$, used by biomass (2 per unit) and by v_{lower}:

$$2(10 - b) = 2b + v_{\text{lower}} \quad \Longrightarrow \quad v_{\text{lower}} = 20 - 4b$$

ATP. Every pyruvate goes round the TCA, which pays best. Each T3P through v_{lower} yields one ATP and then three more:

$$4(20 - 4b) \;\geq\; 4b \quad\Longrightarrow\quad 80 - 16b \geq 4b \quad\Longrightarrow\quad b \leq 4$$

So $b = 4$, and the rest falls out: $v_{\text{zwf}} = 4$, $v_{\text{pgi}} = v_{\text{tpi}} = 6$, $v_{\text{lower}} = 4$, $v_{\text{tca}} = 4$, $v_{\text{ackA}} = 0$, $v_{\text{atpm}} = 0$.

Check it. ATP made $= 4 + 3(4) = 16$; ATP used $= 4b = 16$. Nothing spare, nothing short.

▶ **Why does that step follow?** The branch at pyruvate went *all* to the TCA and none to acetate. Nothing in $S\mathbf{v} = 0$ says it must. What made the split all-or-nothing?

2 · Last step blank — less glucose, not less network

Nothing is deleted. The medium is leaner: glucose now arrives at **5** instead of 10, and every reaction is still there.

Four of the five lines carry straight over with 10 replaced by 5. The ATP line is untouched — one ATP at lower, three round the TCA, four spent per unit of biomass:

$$v_{\text{zwf}} = b \qquad v_{\text{pgi}} = v_{\text{tpi}} = 5 - b \qquad 2(5 - b) = 2b + v_{\text{lower}}$$

$$_____ \;\geq\; 4b$$

▶ $b =$ _____, which is ____**% of the wild type.**

▶ **Why does that step follow?** Halving the glucose did one thing to the answer; deleting the TCA did something else entirely — in class the knockout took b from 4 to 10/3, and v_{ackA} went from carrying nothing to carrying everything. Say in one sentence what is different about the two changes, in terms of where the optimum sits.

3 · More blank — counting what is left over, and proving what kills it

(a) Degrees of freedom, after a deletion. Deleting a reaction removes a **column** of S. Take the network with v_{tca} gone.

Rows: _____ Columns: _____ rank S = _____

$$\dim \text{null}(S) \;=\; r - \text{rank}(S) \;=\; _____$$

▶ **Why does that step follow?** The full network had three directions the stoichiometry could not choose between. Session 3 put **two** null spaces on the board, one for what is conserved and one for what can flow.

Which of the two is this, and what does the *other* one count?

(b) Lethality, proved. In class you named the four deletions that drive b to exactly zero. Naming them is not proving them. For each one, write **the row of S** that forces $b = 0$, and finish the argument in one line.

A row of S is one metabolite's balance: what makes it, minus what uses it, equal to zero. Two of the four need a second row to finish — say which, and which row.

DELETED	THE ROW THAT DOES IT	AND SO
v_{zwf}		
v_{tpi}		
v_{pgi}		
v_{lower}		

▶ **Why does that step follow?** Every one of those arguments used $\mathbf{v} \geq 0$ somewhere, and would collapse without it. Point at the place in one of them where it was used, and say what goes wrong if a flux is allowed to run negative.

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4 · Entirely yours — when does a duplicate change anything?

In class we added one reaction, v_{zwf2}, with exactly the stoichiometry of v_{zwf}, and two things happened at once: the optimum stopped being unique, and the coupling between v_{zwf} and b disappeared. Neither of those is automatic. This section is about when a duplicate does that and when it does nothing at all.

(a) Add instead v_{tca2}, with exactly the TCA's stoichiometry: PYR $\longrightarrow$ 3 CO_2 + 3 ATP. Nothing else changes. What is the optimal b? Which fluxes does flux variability analysis now report as free, and over what range?

(b) Which coupled pairs survive in that network? Compare with what happened in class when v_{zwf} was duplicated, and account for the difference in one sentence about **rows of S**.

(c) Now add v_{ackA2} instead, with exactly ackA's stoichiometry. What does flux variability analysis return this time? Say why the answer is not the same kind as (a).

(d) State the rule you have just found. What has to be true of a reaction before duplicating it can create alternate optima at all?

Before Thursday

Mishra *et al.*, *Nat Biotechnol* 32:1268 (2014) — a load driver. A circuit slows down because something downstream is *reading* it. Read Figure 1 and the rise times in the Online Methods.

Come able to say what the "load" physically is.